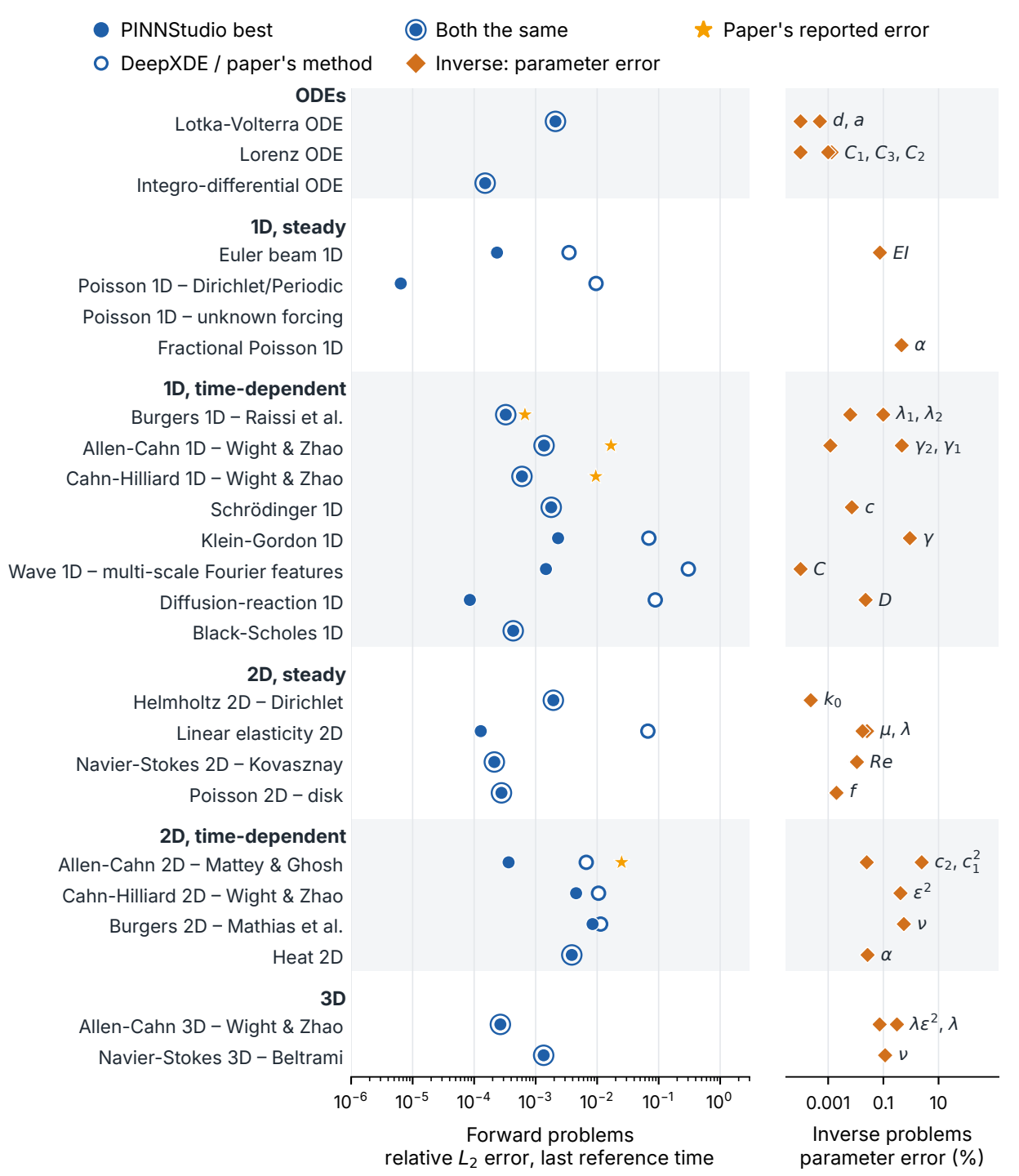
PINNStudio best
Both the same
Paper's reported error
DeepXDE / paper's method
Inverse: parameter error
ODEs
Lotka-Volterra ODE
Lorenz ODE
Integro-differential ODE
1D, steady
Euler beam 1D
Poisson 1D – Dirichlet/Periodic
Poisson 1D – unknown forcing
Fractional Poisson 1D
1D, time-dependent
Burgers 1D – Raissi et al.
Allen-Cahn 1D – Wight & Zhao
Cahn-Hilliard 1D – Wight & Zhao
Schrödinger 1D
Klein-Gordon 1D
Wave 1D – multi-scale Fourier features
Diffusion-reaction 1D
Black-Scholes 1D
2D, steady
Helmholtz 2D – Dirichlet
Linear elasticity 2D
Navier-Stokes 2D – Kovasznay
Poisson 2D – disk
2D, time-dependent
Allen-Cahn 2D – Mattey & Ghosh
Cahn-Hilliard 2D – Wight & Zhao
Burgers 2D – Mathias et al.
Heat 2D
3D
Allen-Cahn 3D – Wight & Zhao
Navier-Stokes 3D – Beltrami
$d$, $a$
$C_1$, $C_3$, $C_2$
$EI$
$\alpha$
$\lambda_1$, $\lambda_2$
$\gamma_2$, $\gamma_1$
$c$
$\gamma$
$C$
$D$
$k_0$
$\mu$, $\lambda$
$Re$
$f$
$c_2$, $c_1^2$
$\varepsilon^2$
$\nu$
$\alpha$
$\lambda\varepsilon^2$, $\lambda$
$\nu$
$10^{-6}$ $10^{-5}$ $10^{-4}$ $10^{-3}$ $10^{-2}$ $10^{-1}$ $10^{0}$
Forward problems
relative $L_2$ error, last reference time
0.001 0.1 10
Inverse problems
parameter error (%)